Coal Mining Midterm

Total points 104/104

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✓ Which of the following type of coal has an estimated heat value ranging between 4,000 and 8,300 BTUs-per-pound. *1/1

- (●) a. Lignite ✓
- ( ) b. Sub-bituminous
- ( ) c. Bituminous
- ( ) d. Anthracite

✓ Carbon exists in the atmosphere, in water, and in rock, generally as _____. *1/1

- ( ) a. carbon monoxide
- (●) b. carbon dioxide ✓
- ( ) c. methane
- ( ) d. coal

✓ The global cycle can be interrupted by the following events EXCEPT ONE: * 1/1

- a. floods
- b. earthquakes
- c. volcanic eruptions
- (●) d. typhoon ✓

✓ Which of the following elements is/are not constituents of a mineral matter? *1/1

- a. Si
- b. Fe
- c. Na
- (●) d. Zn ✓

✓ The _____ that were precipitated during the later stages of coalification are generally found either in cleats and fissures or intruding into a disrupting the original layers of organic material. *1/1

- a. Syngenetic minerals
- (●) b. Epigenetic minerals ✓
- c. Inherent minerals
- d. None

In what coal exploration phase where the objective is to increase drillhole, core density and data accumulation? *1/1

- a. PHASE 1
- b. PHASE 2
- c. PHASE 3
- d. PHASE 4 ✓
- e. PHASE 5

The advantage of this process are the inhomogeneous composition of raw materials, high dust content in the workshop. *1/1

- a. PF Boiler Process
- b. COREX Process
- c. Dry Process ✓
- d. Wet Process

In coal sampling, the sampling implement must intersect the stream of material at a relative velocity. *1/1

- True
- False ✓

✓ Which of the following type of coal produces a long, yellow flame and gives off a suffocating odor when burns? *1/1

- a. lignite
- b. sub-bituminous
- (●) c. bituminous ✓
- d. peat
- e. anthracite

✓ When found in sufficient quantities, such liptinitic-rich sediments are known as _____ (deeper water coals). *1/1

- (●) a. Sapropelic coals ✓
- b. Cannel coals
- c. Boghead coals
- d. Humic coals

✓ Which of the following type of coal can be sourced majority at the United States of America? *1/1

- a. Lignite
- b. Sub-bituminous
- c. Bituminous ✓
- d. Peat
- e. Anthracite

✓ In what coal exploration phase where the objective is to confirm and expand existing date thru field recon for use in the initial acquisition of property control? *1/1

- a. PHASE 1
- b. PHASE 2 ✓
- c. PHASE 3
- d. PHASE 4
- e. PHASE 5

✓ Which of the reserved confidence level is one for which tonnage and grade are computed partly from specific measurements, production data, projections for reasonable geologic evidence? *1/1

- a. Speculative-Hypothetical
- b. Inferred
- c. Indicated ✓
- d. Measured

✓ The buried, accumulated organic matter undergoes further decay by bacteria and after thousand of years, _____ forms? *1/1

- a. peat
- b. kerogen ✓
- c. lignite
- d. coal

✓ Which of the following maceral group predominantly aliphatic in molecular structure and low in density? *1/1

- a. Vitrinite maceral group
- b. Inertinite maceral group
- c. Liptinite maceral group ✓
- d. None

✓ The higher the grade of coal, the more versatile its uses. * 1/1

- (●) True ✓
- ( ) False

✓ High sulfur content in coal will improve to the heating value of coal yet heating and oxidizing it leads to corrosion. *1/1

- ( ) True
- (●) False ✓

✓ Which of the following maceral group is more structurally ordered and aromatic? *1/1

- ( ) a. Vitrinite maceral group
- (●) b. Inertinite maceral group ✓
- ( ) c. Liptinite maceral group
- ( ) d. None

✓ In coal sampling, the smaller the number of increments collected, the better the precision of the sampling. *1/1

- ( ) True
- (●) False ✓

✓ Coal-derived fly-ash is used as an additive in concrete mixes as a substitute for Portland cement. *1/1

- (●) True ✓
- ( ) False

✓ Coal is also known as _____. * 1/1

- ( ) a. Black gem
- ( ) b. Black diamond
- (●) c. Black gold ✓
- ( ) d. Black crystal

✓ Which of the following maceral group are grey to pale grey in the mid bituminous rank? *1/1

- ( ) a. Vitrinite maceral group
- (●) b. Inertinite maceral group ✓
- ( ) c. Liptinite maceral group
- ( ) d. None

✓ Which of the following processes used coal undersized ore as initial feed? *1/1

- ○ a. PF Boiler Process
- ◉ b. COREX Process ✓
- ○ c. Dry Process
- ○ d. SASOL HTFT Process

✓ Which of the following processes is a smelting reduction process created by Siemens VAI as a more environmentally friendly alternative to the blast furnace? *1/1

- ○ a. PF Boiler Process
- ◉ b. COREX Process ✓
- ○ c. Dry Process
- ○ d. SASOL HTFT Process

✓ _____ refers to that portion of the organic matter which is driven off as a gas. *1/1

- ○ a. Fixed carbon
- ◉ b. Volatile matter ✓
- ○ c. Calorific value
- ○ d. Agglomerating character

✓ Which of the following maceral group is typically white in color? * 1/1

- ○ a. Vitrinite maceral group
- ◉ b. Inertinite maceral group ✓
- ○ c. Liptinite maceral group
- ○ d. None

✓ The carbon atoms will form chains rather than rings as in oil or gas. * 1/1

- ○ True
- ◉ False ✓

✓ Volatile matter yield and _____ then show a marked decrease with further coalifi *1/1

- ○ a. oxygen
- ◉ b. hydrogen ✓
- ○ c. nitrogen
- ○ d. carbon

✓ The mineral code that have been adopted by the Australian Mining Industry *1/1

- a.PMRC
- b.CRISSCO
- c.SAMRECE
- d.PERC
- (●) e.JORC ✓

✓ Which of the following type of coal has a high percentage of fixed carbon and a low percentage of volatile matter? *1/1

- a. Lignite
- b. Sub-bituminous
- c. Bituminous
- d. Peat
- (●) e. Anthracite ✓

✓ Which of the reserve confidence level is one for which quantitative estimates are based largely on broad knowledge of geologic character of deposit? *1/1

- ○ a. Speculative-Hypothetical
- ◉ b. Inferred ✓
- ○ c. Indicated
- ○ d. Measured

✓ Which of the following type of coal can be subdivided into low-volatile medium-volatile and high-volatile groups? *1/1

- ○ a. Lignite
- ○ b. Sub-bituminous
- ◉ c. Bituminous ✓
- ○ d. Peat
- ○ e. Anthracite

✓ Which of the reserve confidence level is one for which tonnage is computed from dimension measured in outcrop, trenches, drill holes for which the grade is computed from detailed sampling. *1/1

- a. Speculative-Hypothetical
- b. Inferred
- c. Indicated
- d. Measured ✓

✓ This process utilized the vast deposits of lower grade (high ash) bituminous coal by employing indirect liquefaction via the combination of a lurgi fixed-bed gasifier in conjunction with the with the Fischer Tropsch process. *1/1

- a. PF Boiler Process
- b. COREX Process
- c. Dry Process
- d. SASOL HTFT Process ✓

✓ Which of the following processes where the pellets are bake hard in a pre-heating grate by means of hot gasses from the kiln? *1/1

- a. PF Boiler Process
- b. COREX Process
- c. Dry Process ✓
- d. Wet Process

✓ Coal is facing increasing competition from other fuels, mainly natural gas. *1/1

- True ✓
- False

✓ Which of the following maceral group is typically white in color? * 1/1

- a. Vitrinite maceral group
- b. Inertinite maceral group ✓
- c. Liptinite maceral group
- d. None

✓ _____ refers to a standard test made to determine whether a pulverized sample of coal will or will not fuse a form a solid button at an elevated temperature. *1/1

- a. fixed carbon
- b. volatile matter
- c. calorific value
- d. agglomerating character ✓

✓ Which of the following maceral group is grey in color? * 1/1

- a. Vitrinite maceral group ✓
- b. Inertinite maceral group
- c. Liptinite maceral group
- d. None

✓ These are all boring specifications EXCEPT ONE: * 1/1

- a. Depth of holes
- b. Sample preservation
- c. Surface water ✓
- d. Groundwater
- e. None of these

✓ The dry process uses more energy in grinding, but less in the kiln. * 1/1

- (●) True ✓
- ( ) False

✓ The mineral code that have been adopted by the Philippine Mining Industry. *1/1

- (●) a. PMRC ✓
- ( ) b. CRISSCO
- ( ) c. SAMRECE
- ( ) d. PERC
- ( ) e. JORC

✓ The blast furnace processes the most mature and efficient technology for producing hot metal for steelmaking. *1/1

- (●) True ✓
- ( ) False

✓ These are all factors to consider for usage of coal EXCEPT ONE: * 1/1

- a. heating value
- (●) b. mechanical flexibility ✓
- c. sulfur and other impurities
- d. nono of these

✓ Pulverized coal or fuel (PF) combustion is the most commonly used technology for coal-based power generation, accounting for approximately 40% of the total electricity supply. *1/1

- (●) True ✓
- False

✓ This uses a rotary-type of drill machine and drilling fluids in conjunction. * 1/1

- a. Rotary Drilling
- (●) b. Conventional Coring Drilling ✓
- c. Wire-line Coring Drilling
- d. Percussion Drilling

✓ Higher calorific value means low efficiency as ash and moisture is low. * 1/1

- ○ True
- ◉ False ✓

✓ This is applicable for shallow overburden where high penetration rate needed Rotary Drilling. *1/1

- ◉ a. Rotary Drilling ✓
- ○ b. Conventional Coring Drilling
- ○ c. Wire-line Coring Drilling
- ○ d. Percussion Drilling

✓ Metallurgical coal or coking coal is vital for sustainable steel manufacturing. *1/1

- ◉ True ✓
- ○ False

✓ _____ is the heat produce by combustion of coal. * 1/1

- a. Fixed carbon
- b. Volatile matter
- (●) c. Calorific value ✓
- d. Agglomerating character

✓ Which of the following type of coal is mostly associated with home heating. *1/1

- a. Lignite
- b. Sub-bituminous
- c. Bituminous
- (●) d. Anthracite ✓

✓ In what coal exploration phase where the objective is to obtain more detailed data and commit the reserve for the final mine feasibility study. *1/1

- a. PHASE 1
- b. PHASE 2
- c. PHASE 3
- d. PHASE 4
- (●) e. PHASE 5 ✓

✓ The role of coal in the processes leading to the production of cement and concrete therefore includes the following EXCEPT ONE: * 1/1

- ○ a. Coal-derived power plant ash (fly-ash and bottom ash) is widely used as a constituent raw material
- ○ b. Pulverized coal is used as a fuel in rotary kilns
- ○ c. Coal-derived fly-ash is used as an additive in concrete mixes as a substitute for Portland cement
- ○ d. Some ash products are used for the production of lightweight aggregates for building materials.
- ◉ e. None of these ✓

✓ The lower the grade of coal, the cleaner it burns. * 1/1

- ○ True
- ◉ False ✓

✓ The dry process has low r overheads than the wet process. * 1/1

- ○ True
- ◉ False ✓

✓ Which of the following processes is also called as early peat accumulation? *1/1

- (●) a. Biochemical degradation ✓
- ( ) b. Oxidation
- ( ) c. Putrefaction
- ( ) d. Geochemical coalification

✓ Cement is a man-made power which, when mixed with water and aggregates, produces concrete. *1/1

- (●) True ✓
- ( ) False

✓ Which of the following type of coal is dense, shiny black, and homogeneous with no marks of layers? *1/1

- ( ) a. Lignite
- ( ) b. Sub-bituminous
- ( ) c. Bituminous
- ( ) d. Peat
- (●) e. Anthracite ✓

✓ In coal-fired power generation, efficiency is a key performance parameter. * 1/1

- (●) True ✓
- ( ) False

✓ Oxyfuel combustion has been also proposed as one transition technology for addressing CO2 issues relating to existing power stations. *1/1

- (●) True ✓
- ( ) False

✓ Which of the following type of coal has an estimated heat value below 4,000 Btu/lb. *1/1

- ( ) a. Lignite
- ( ) b. Sub-bituminous
- ( ) c. Bituminous
- (●) d. Peat ✓
- ( ) e. Anthracite

✓ Natural gas can be injected into tl1e blast furnace as a replacement for coal. *1/1

- (●) True ✓
- ( ) False

✓ _____ refers to that portion of the organic matter which is driven off as a gas. *1/1

- ( ) a. Fix carbon
- (●) b. Volatile matter ✓
- ( ) c. Calorific value
- ( ) d. Agglomerating character

✓ _____ the carbon residue of the organic material that involves the removal of volatile constituents (as a gas) of the organic matter by heat without combustion. *1/1

- (●) a. Fixed carbon ✓
- ( ) b. Volatile matter
- ( ) c. Calorific value
- ( ) d. Agglomerating character

✓ Carbon dioxide emissions from coal-fired power generation are much higher compared to those from alternative fuels. *1/1

- (●) True ✓
- ( ) False

✓ Minerals that were deposited at the same time as peat formation are termed _____ *1/1

- (●) a. syngenetic minerals ✓
- ( ) b. epigenetic minerals
- ( ) c. inherent minerals
- ( ) d. none

✓ Which of the following coal that are enriched in algal matter? * 1/1

- ( ) a. Sapropelic coals
- ( ) b. Cannel coals
- (●) c. Boghead coals ✓
- ( ) d. Humic coals

✓ Which of the following type of coal has a low heating value and a high moisture content? *1/1

- (●) a. Lignite ✓
- ( ) b. Sub-bituminous
- ( ) c. Bituminous
- ( ) d. Peat
- ( ) e. Anthracite

✓ Which of the following type of coal was formed during the diagenetic phase? *1/1

- ( ) a. Lignite
- ( ) b. Sub-bituminous
- ( ) c. Bituminous
- (●) d. Peat ✓
- ( ) e. Anthracite

✓ This Method uses a pump or compressor to circulate a fluid (air, water or drilling mud) through the rods to cool and lubricate the bit and flush the cuttings upward, outside the drill rod to the surface. *1/1

- (●) a. Rotary Drilling ✓
- ( ) b. Conventional Coring Drilling
- ( ) c. Wire-line Coring Drilling
- ( ) d. Percussion Drilling

✓ In determining the ranks of coal, mineral matter is excluded from the analysis as it in no way reflects the degree of metamorphism of the coal. *1/1

- (●) True ✓
- ( ) False

✓ Which of the following type of coal is geologically young coal with 35%-45% carbon content? *1/1

- ( ) a. Lignite
- (●) b. Sub-bituminous ✓
- ( ) c. Bituminous
- ( ) d. Peat
- ( ) e. Anthracite

✓ In coal sampling, the mass of each primary increment needs to be large enough to ensure that variance resulting from inherent heterogeneity is not a major component of the total sampling, preparation and testing error. *1/1

- (●) True ✓
- ( ) False

✓ These are all condition necessary for coal formation EXCEPT ONE: * 1/1

- ( ) a. Swaps or marsh environment and climate favorable to plant growth;
- ( ) b. Some subsidence (sinking) of the area during accumulation of vegetal debris, or compaction of deposited plant materials, permitting further accumulation;
- ( ) c. Sufficiently dry conditions to permit exclusion of air from much of the vegetal material before it decays proximity to the sea or a subsiding area so that vegetal material can be buried by sediments.
- (●) d. Site of accumulation such that removal by erosion does not subsequently occur ✓

✓ Minerals that were deposited at the same time as peat formation are termed _____. *1/1

- (●) a. Syngenetic minerals ✓
- ( ) b. Epigenetic minerals
- ( ) c. Inherent minerals
- ( ) d. None

In what coal exploration phase where the objective is to identify exploration target? *1/1

- (selected) a. PHASE 1 ✓
- b. PHASE 2
- c. PHASE 3
- d. PHASE 4
- e. PHASE 5

_____ defined as small, microscopically identifiable units of altered organic plant remains seen under the petrographic microscope. *1/1

- (selected) a. Maceral ✓
- b. Micromacerals
- c. Lithotypes
- d. Microlithotypes

✓ Which of the following type of coal is used for the production of steel and cement, electricity generation and coke production? *1/1

- a. Lignite
- b. Sub-bituminous
- c. Bituminous ✓
- d. Peat
- e. Anthracite

✓ Which of the following type of coal has been estimated nearly half of the world's proven coal reserves are made up? *1/1

- a. Lignite ✓
- b. Sub-bituminous
- c. Bituminous
- d. Peat
- e. Anthracite

✓ Which of the following processes where raw material are grind into slurry after mixing with water and then feeding them into kiln for drying and calcination and finally forming clinker? *1/1

- a. PF Boiler Process
- b. COREX Process
- c. Dry Process
- d. Wet Process ✓

✓ Graphite can be found in association with some anthracite deposits. * 1/1

- True ✓
- False

✓ Where the spores, pollens and leaf cuticles concentrate in such sapropelic coals, these are specifically known as _____. *1/1

- a. Sapropelic coals
- b. Cannel coals ✓
- c. Boghead coals
- d. Humic coals

✓ Mineral matter is generally an undesirable constituent and is the source of variety of problems associated with the use of coal. *1/1

- (●) True ✓
- ( ) False

✓ In coal sampling, the width of the sampling implements or cutter must be at least three times wider than the nominal top size (as opposed to top size) of the coal being sampled. *1/1

- (●) True ✓
- ( ) False

✓ The heat consumption of this process is too high, usually between 5234-490 J/kg and the consumption of ball mill vulnerable parts is also large. *1/1

- ( ) a. PF Boiler Process
- ( ) b. COREX Process
- ( ) c. Dry Process
- (●) d. Wet Process ✓

✓ Which of the following maceral group has a well-preserved cellular structure and appears hard (high relief), high reflective, and very bright white to brassy yellow in color? *1/1

- a. Vitrinite maceral group
- (●) b. Inertinite maceral group ✓
- c. Liptinite maceral group
- d. None

✓ Coal-derived power plant ash (fly-ash and bottom ash) is widely used as a constituent raw material. *1/1

- (●) True ✓
- False

✓ Coal-based power generators are attempting to reduce the environmental impact by increasing the efficiency of established pulverized coal-fired technology. *1/1

- (●) True ✓
- False

✓ Which of the following processes used pulverized coal as initial feed? * 1/1

- (●) a. PF Boiler Process ✓
- ( ) b. COREX Process
- ( ) c. Dry Process
- ( ) d. SASOL HTFT Process

✓ Which of the following mineral matter on coal seams are int eh form of fine particles? *1/1

- ( ) a. Clay
- (●) b. Quartz ✓
- ( ) c. Sand and silt
- ( ) d. Sulfur

✓ Which of the followiing maceral group is moderately aromatic with significant aliphatic chains in the bituminous range of rank? *1/1

- (●) a. vitrinite maceral group ✓
- ( ) b. inertinite maceral group
- ( ) c. liptinite maceral group
- ( ) d. none

In what coal exploration phase where the objective is to acquire control of key tracts within the target area and conduct initial acquisition of property? *1/1

- a. PHASE 1
- b. PHASE 2
- c. PHASE 3 ✓
- d. PHASE 4
- e. PHASE 5

Which of the following type of coal is the fastest growing market for coal that is supplying heat for industrial purposes. *1/1

- a. Lignite
- b. Sub-bituminous
- c. Bituminous ✓
- d. Anthracite

Clinker is made in the kiln at temperatures of 1650°C. * 1/1

- True
- False ✓

✓ Which of the following type of coal is geologically young coal with 25%-35% carbon content? *1/1

- (●) a. Lignite ✓
- ( ) b. Sub-bituminous
- ( ) c. Bituminous
- ( ) d. Peat
- ( ) e. Anthracite

✓ These are all factors to consider in selection of drilling methods EXCEPT ONE: *1/1

- ( ) a. Hole size
- ( ) b. Drilling cost
- (●) c. Effectivity of method ✓
- ( ) d. Efficiency of the method
- ( ) e. None of these

This method is preferred over conventional system due to the time saved in retrieving the core sample. *1/1

- a. Rotary Drilling
- b. Conventional Coring Drilling
- c. Wire-line Coring Drilling ✓
- d. Percussion Drilling

No two coals are alike * 1/1

- True ✓
- False

Which of the following mineral matter on coal seams are usually present in swaps and are taken by plants? *1/1

- a. Clay
- b. Quartz
- c. Sand and silt
- d. Sulfur ✓

✓ Which of the following mineral matter on coal seams are similar to those found in rocks in the coal-bearing sequence? *1/1

- (●) a. Clay ✓
- ( ) b. Quartz
- ( ) c. Sand and silt
- ( ) d. Sulfur

✓ Coal was formed from accumulation of plant material buried beneath the surface that has undergone complete oxidation. *1/1

- ( ) True
- (●) False ✓

✓ Which of the following maceral group is predominantly aliphatic? * 1/1

- ( ) a. Vitrinite maceral group
- ( ) b. Inertinite maceral group
- (●) c. Liptinite maceral group ✓
- ( ) d. None

✓ 61. Which of the following maceral group are brown to purple in color in sub-bituminous? *1/1

- a. Vitrinite maceral group
- b. Inertinite maceral group
- (●) c. Liptinite maceral group ✓
- d. None

✓ Pulverized coal is used as a fuel in rotary kilns. * 1/1

- (●) True ✓
- False

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